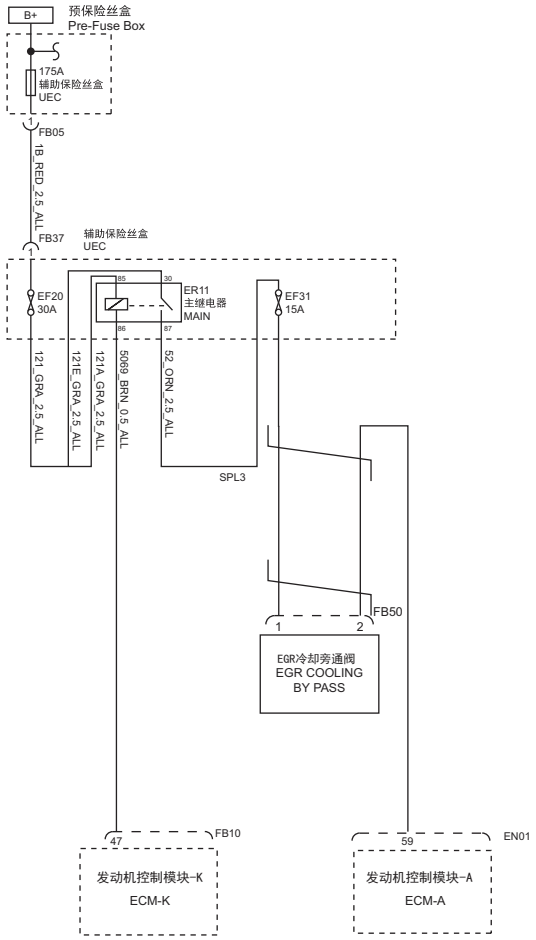
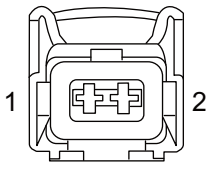
P00E3

Description of fault diagnosis codes
P00E3-11：Short circuit to ground error of ECB Valve
P00E3-12：Short circuit to battery error of ECB Valve
P00E3-13：open load error for ECBVlv
P00E3-4B：Over temperature error on powerstage for EGR cooler bypass valve
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • EGR cooler bypass valve fault
Description of circuit diagram
The exhaust gas recirculation cooling bypass-valve is used to cool the exhaust gas recirculation valve by controlling the coolant circulation, the main relay provides 12v voltage to the exhaust gas recirculation cooling bypass-valve, and the engine control module provides grounding control to the low reference voltage circuit of exhaust gas recirculation cooling bypass-valve.

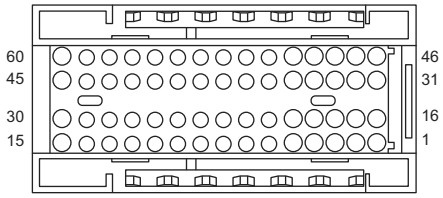
Circuit diagram



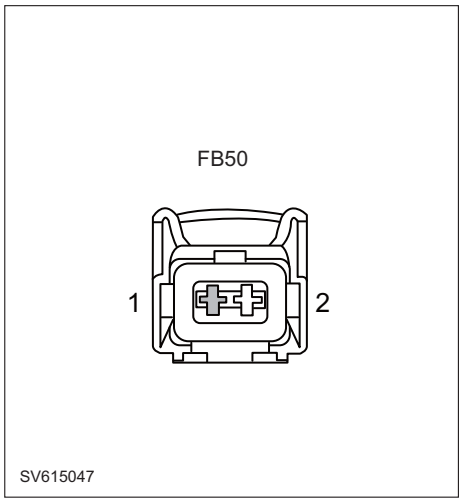
FB50

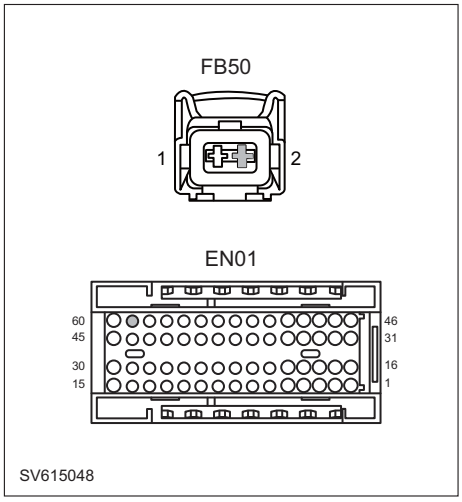


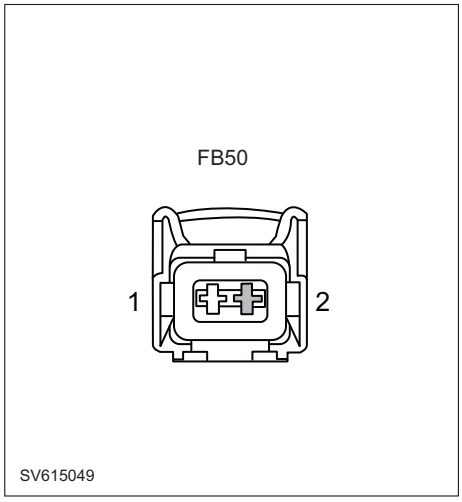
EN01



SV615046

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse EF31 for damage. 2 Check battery capacity. 3 Check the engine control module and the harness plug of EGR cooler bypass valve for damage, poor contact, aging and looseness. 4 Check whether the result is normal.
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for POOE3?
		Yes → Reference: DTC Summary List Index.
No ↓		
Step 3	Check the power supply circuit of EGR bypass valve.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of EGR bypass valve. 3 Turn the ignition switch to ON position. 4 Measure the voltage between the harness plug of EGR cooler bypass valve and ground. Measure circuit: the voltage between FB50 harness plug terminal 1 and ground. **Standard value: 10~14V** 5 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		

Step 4	Check the control circuit of EGR cooler bypass valve (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Measure the resistance of harness plug between EGR cooler bypass valve and engine control module. Measure circuit: the resistance between FB50 plug terminal 2 and EN01 plug terminal 59. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the control circuit of EGR cooler bypass valve (short circuit).
	1 Measure the resistance between the harness plug of EGR cooler bypass valve and ground. Measure circuit: the resistance between FB50 plug terminal 2 and ground. **Standard value: ∞** 2 Measure the voltage between the harness plug of EGR cooler bypass valve and ground. Measure circuit: the voltage between FB50 plug terminal 2 and ground. **Standard value: 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the control circuit of EGR bypass valve.
	1 Connect the harness plug of EGR cooler bypass valve and engine control module. 2 Turn the ignition switch to ON position. 3 Measure the voltage between the harness plug terminal of EGR cooler bypass valve and ground. Measure circuit: the voltage between FB50 plug terminal 2 and ground. **Standard value: 3~4V** 4 Check whether the result is normal?
	No → Replace engine control module.
Yes ↓	
Step 7	Check the EGR cooler bypass valve(components).
	1 Measure the resistance among the terminals of EGR cooler bypass valve. Measure circuit: Check the resistance between terminal 1 and terminal 2 of EGR cooler bypass valve. **Standard value: 23~24Ω** 2 Check whether the result is normal?
	No → Replace the EGR cooler bypass valve.
Next ↓	
Step 8	Complete the check and confirm whether the system works normally nor not.